Rapport annotation pipeline

This pipeline outlines the procedures for collecting and analyzing rapport in dyadic interactions using video data.

To start with the data structure, the video dataset is hierarchically structured around 22 dyads, each recorded over the course of three weeks. In each week, dyads participated in two interaction phases: a collaboration phase, which involved a structured, task-oriented activity, and a free interaction phase, which captured spontaneous, natural conversation. Each phase lasted five minutes and was segmented into ten video clips of 30 seconds each. As a result, there are 20 clips per week per dyad, totaling 60 clips per dyad over the three weeks. Across all 22 dyads, the dataset comprises a total of 1,320 video clips.

Each clip is labeled using the format dyad[N]_w[week]_[phase]_[clip number]—for example, dyad1_w1_collab_0 refers to the first collaboration clip of week one for dyad 1. This naming allows for systematic tracking of the clips across dyads, weeks, and phases.

For annotation, the 30-second clips are randomly assigned into batches of five, which are presented to annotators as experiments hosted on **Gorilla** and **Prolific**.

In this pipeline, we will first describe how to set up and run a rapport annotation experiment using Gorilla for task and experiment design, and Prolific for participant recruitment. This pipeline will guide you through the setup process on both platforms and explain how they work together.

The final section will be on data extraction and post-collection annotation processing.

A) How to set up a project on Gorilla?

Gorilla projects consist of two main components: the **Task** and the **Experiment**. The task defines what participants will see and do, while the experiment controls how the task is delivered and how participants are recruited. In our case, participants will be recruited through Prolific.

Structure of the Task

The task is divided into the following components:

1. Instructions
2. Practice (Training)
3. Main Task
4. Debrief + Questionnaire

What Does the Task Involve?

1) Instruction Screen

Participants are first introduced to the concept of rapport, or connivence in French. They are given a definition and asked to focus on behavioral cues to evaluate it.

Definition of Rapport:

> *Le terme "connivence" décrit un certain type d'interaction entre deux personnes. Ce type d'interaction est souvent caractérisé par des expressions telles que « nous avons vraiment accroché » ou « nous avons été sur la même longueur d'ondes ». Des termes comme "harmonieux", ou "en phase" décrivent des interactions ayant une connivence élevée.*

Participants are then given instructions about their role in the study:

> *Votre tâche est d'évaluer le niveau de connivence pour une série de clips vidéo, chacune montrant une interaction entre 2 personnes. Vous allez voir des clips d'interactions de binômes différents, et l'ordre des clips a été aléatoirisé. Donc chaque tâche que vous faites vous montrera une tranche qui n'a rien à voir avec les autres tranches de vidéo que vous annoterez.*
>
> *On vous demande d'évaluer le niveau de connivence entre les deux personnes dans chaque vidéo de 30 secondes sur une échelle de 1 à 7, où: 1 = absence totale de connivence et 7 = connivence maximale.*
>
> *Vous devez également évaluer votre confiance dans votre annotation sur une échelle de 1 à 7, où: 1 = aucune confiance et 7 = confiance totale*

2) Practice Phase (Training Videos)

Before the real annotations, participants are shown 3 training videos representing different rapport levels:

- One low rapport video
- One neutral rapport video
- One high rapport video

3) Main Evaluation Task

After training, participants will evaluate 5 real video clips, each 30 seconds long, using the same rapport and confidence rating scales.

4) Final Questionnaire

At the end of the task, participants must complete a brief open-ended questionnaire. They are asked to explain what factors influenced their evaluations for low, neutral, and high rapport clips.

Instructions for setting up a project

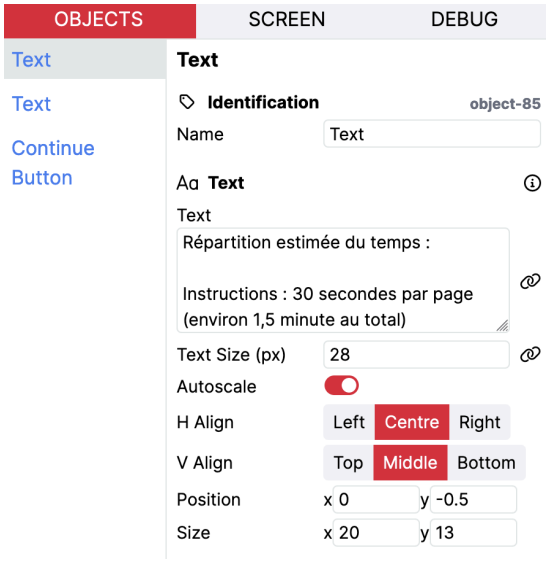
1. Create a New Project
    - Click on "Create New Project."
    - Rename the project to “Group [Experiment Number]” (e.g., *Group 1, Group 2*, etc.).
2. Create a New Task
    - Inside the project, click on "Create New" → select "Task Builder."
    - Choose "Clone Existing."
    - Select the task from Group 1.
    - Click on “Rapport.”
    - Click “Create.”

The cloned task contains four displays. The first display is titled "Instructions" and consists of four sequential instruction screens.

Each screen presents information to the participant. A “Continue” button appears in the bottom-right corner after a 10-second delay, allowing participants to proceed. This delay is time-triggered, ensuring participants have time to read before continuing.

The instruction content is added as "Text Objects." To modify or add text:

- Click “Edit” on the task.
- Navigate to the screen you want to change.
- Click “Add Object” → select “Text.”
- A window will appear on the right side of the page.
- Enter your content in Markdown format in the provided text box.

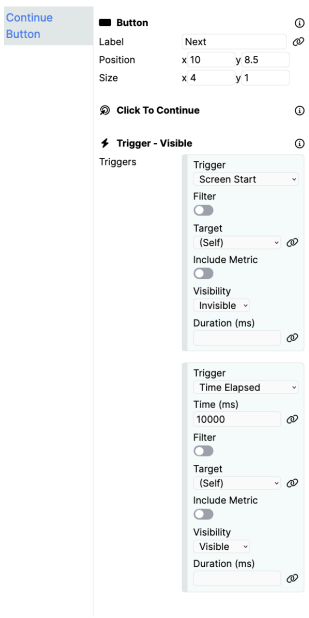


To allow participants to progress through the instruction screens, you'll need to add a Continue Button Object that appears 10 seconds after the screen loads.

- In the Task Builder, go to the instruction screen where you want to add the button.
- Click "Add Object" → select "Continue Button."
- A user-friendly settings panel will appear on the right side of the screen.

To configure the Trigger Settings, you'll need to set up when the button becomes visible:

- Trigger – Screen Start:
    - Metric: invisible
    - This ensures the button is not visible when the screen first loads.
- Trigger – Time Elapsed:
    - Metric: 10000 ms (10 seconds)
    - Action: visible
    - This ensures the button appears after 10 seconds, allowing the participant to proceed.



B) How to set up a study on Prolific?

Section 1: Data collection type

Select “external study link”

Section 2: Study details

a) Study name: Name it "étude de rapport [experiment number]”
b) Internal study name: leave it blank
c) Study description: "Dans cette étude on vous demandera d'évaluer la connivence entre deux personnes”
d) Data collection type: select annotation
e) Device requirements: leave blank
f) Labels: none

Section 3: Data collection

a) What’s the URL of the study? → Put in the URL you get from Gorilla
b) Recording Prolific IDs → leave it on default, that is “URL parameters”
c) Completion paths → leave it on default, that is “manually review"
    i) Here, copy the URL that appears underneath “Redirect URL”. This will be used on Gorilla.

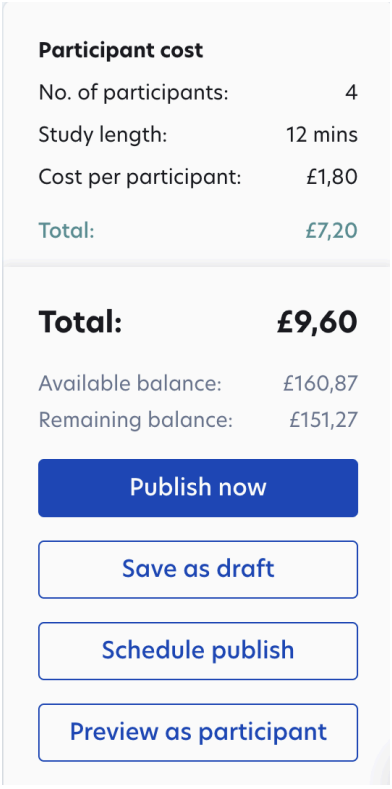
Section 4: Recruit participants

a) Source → select "Find new participants on Prolific"
b) Participants → enter the number "4"
c) Credentials → select no
d) Screening → select "use a saved screener set". Here select the screener that is created under the name "Native French speakers living in France metropolitaine". It will print the preselected country of residence as France and first language as French.
e) Submissions → Select "once"

Section 5: Cost

In France, we need to pay the equivalent of a SMIC corresponding to the time it takes for participants to complete our task. Our task takes 12 minutes and we pay £1.80 per participant.

Total cost of the experiment looks as the following:



When we click on the publish now, we run one experiment where the same 5 videos are shown to 4 different participants, i.e annotators. This experiment is linked to a Gorilla project with the corresponding name (For example if the Gorilla project is Group 28, then the Prolific study name is étude de rapport 28). As it is seen on the image above, in addition to the total costs per participant, the platform charges an additional £2,40 per experiment.

C) Post-collection annotation processing

After data collection via Gorilla, we follow several steps to extract, organize, and analyze the annotations:

1. Data Extraction

Gorilla exports a .csv file containing metadata (e.g., task start time, video URL, reaction times). From this export, we extract the following key columns into a separate .csv file:

- Response (rapport score and confidence)
- Participant ID
- Video URL (to link the response to the correct clip)

2. Data Aggregation

We combine responses across all participants for a given dyad, aggregate data across multiple collection weeks, and then merge these into a single dataframe.

3. Outlier Removal

We implement an outlier removal procedure before applying the weighted majority rule. We calculate the mean of all four ratings, then identify and discard the single rating that deviates most from this mean (i.e., has the largest absolute difference). We then apply the weighted majority rule to the remaining three ratings to determine the final consolidated score.

4. Consolidated Score Calculation

To derive consolidated rapport scores from the annotations, we use a weighted averaging approach that accounts for annotator reliability differences. First, we calculate inter-rater reliability using Krippendorff's alpha both at the overall dataset level and for individual video segments to assess data quality and identify segments with poor annotator agreement. This is to see whether some slices should be excluded or interpreted more carefully.

We then calculate a consolidated score using the **weighted majority rule** with reverse bias correction. That is, each annotator's rating is assigned a weight of $1/\text{Freq}_i(k)$, where $\text{Freq}_i(k)$ represents how frequently annotator i has used that specific rating category across all their annotations (annotators who overuse certain ratings receive lower weights for those categories). We calculate the total weighted score for each possible rating category by summing the weights of all annotators who selected that category, then assigning the final rapport score k^* as the category with the highest total weighted score.